

Few dates t_{DEM}

Surface & Bed DEMs

$$z_s(x, y, t_{\text{DEM}}), z_b(x, y)$$

Force Balance · Elmer/Ice

Full Stokes solved at each t_{DEM}

Empirical relationships

$$\tau_b(t_{\text{DEM}}) \approx g(H(t_{\text{DEM}}))$$

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$$u_{\text{def}}(t_{\text{DEM}}) \approx h(H^{n+1}(t_{\text{DEM}}))$$

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Many dates t_{Stake}

Field Observations

In-situ stake readings

$$H(t_{\text{Stake}})$$

Ice thickness

$$u_{\text{surf}}(t_{\text{Stake}})$$

Surface velocity

TEMPORAL INTERPOLATION via empirical SIA relationships

$$\tau_b(t_{\text{Stake}}) \approx g(H(t_{\text{Stake}}))$$

Basal shear stress

$$u_{\text{def}}(t_{\text{Stake}}) \approx h(H^{n+1}(t_{\text{Stake}}))$$

Deformation velocity

$$u_{\text{bed}}(t_{\text{Stake}}) = u_{\text{surf}}(t_{\text{Stake}}) - u_{\text{def}}(t_{\text{Stake}})$$

Basal sliding velocity

$$\text{Friction Law : } \tau_b = f(u_{\text{bed}})$$

Fit $\tau_b(t_{\text{Stake}})$ vs $u_{\text{bed}}(t_{\text{Stake}})$ — Weertman- & Lliboutry-type laws

Input data Model output Observed timeseries Interpolated timeseries Final result